

Domain Transform Investigation– Basis Images, Energy Compaction, and Filtering

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Part 1: Basis Images

Q: Which basis images correspond to low frequencies and which to high frequencies? What is the relationship between the indices k_1 and k_2 and the frequency content?

A:

I think that low-frequency basis images are located near the **top-left corner** of the grid, where both indices k_1 and k_2 are small (e.g., $(0, 0)$, $(0, 1)$, $(1, 0)$). These images exhibit **smooth, gradual variations** in intensity and represent the coarse or overall structure of an image.

High-frequency basis images are located near the **bottom-right corner** of the grid, where both indices k_1 and k_2 are large (e.g., $(7, 7)$, $(7, 6)$, $(6, 7)$). These images exhibit **rapid oscillations** and represent fine details, edges, and texture.

The indices k_1 and k_2 directly correspond to the **frequency** in the vertical and horizontal directions respectively:

- k_1 controls the **vertical frequency** (oscillations along rows).
- k_2 controls the **horizontal frequency** (oscillations along columns).
- Larger indices $\Rightarrow$ higher frequency $\Rightarrow$ more rapid spatial variations.

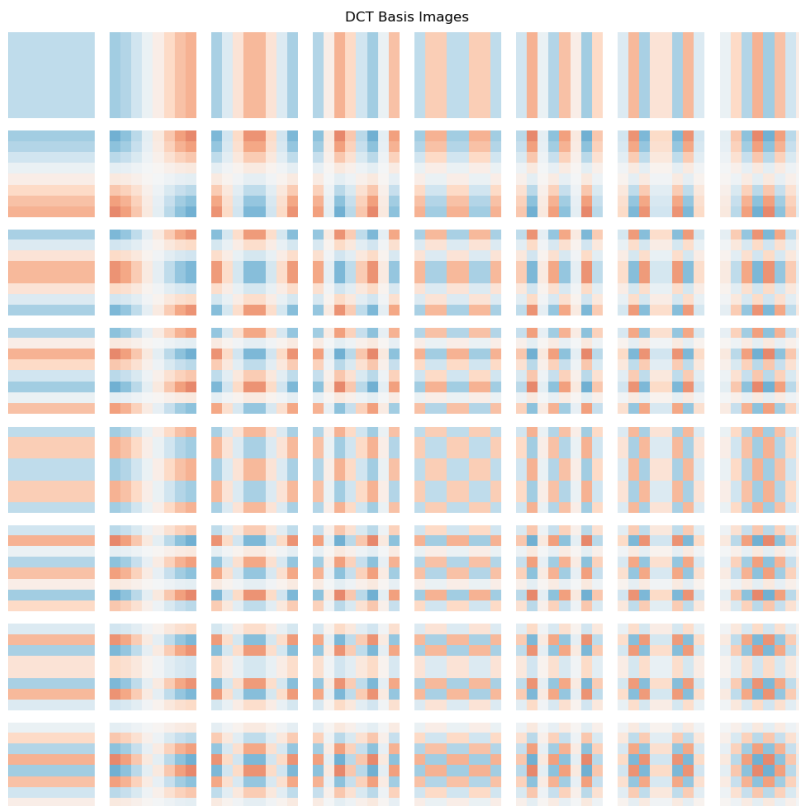


Figure 1: DCT Basis Images ($N=8$).

Part 2: 2D Transform Implementation and Separability

Q: Why is the `norm='ortho'` parameter critical? How does it help preserve the properties of a unitary transform?

A:

The `norm='ortho'` parameter is critical because it ensures the DCT transform is **orthonormal (unitary)**, which provides several essential mathematical properties:

1. Energy Preservation (Parseval's Theorem):

$$\sum_{i,j} |x_{i,j}|^2 = \sum_{k_1,k_2} |X_{k_1,k_2}|^2$$

The energy of the signal is preserved in the transform domain.

2. Perfect Reconstruction:

The transform is exactly invertible:

$$x = \text{IDCT}(\text{DCT}(x))$$

as evidenced by the MSE

$$\approx 5.78 \times 10^{-32},$$

which is essentially zero (limited only by floating-point precision).

3. Exact Inverse:

The inverse transform is simply the transpose (or adjoint) of the forward transform:

$$\text{DCT}^{-1} = \text{DCT}^T.$$

Without `norm='ortho'`, the transform would include scaling factors, requiring manual normalization for perfect reconstruction and preventing energy conservation.

Part 3: Energy Compression and Reconstruction

Q: In the case of keeping only 5% of the coefficients, how is the visual quality of the image? What does this phenomenon teach us about the efficiency of DCT in compression systems like JPEG?

A:

When keeping only **5% of the largest coefficients**, the reconstructed image maintains **good visual quality** with:

- Preserved overall structure and main features.
- Slightly blurred fine details and edges.
- PSNR value typically around **30–35 dB**, which is considered acceptable quality.

This phenomenon demonstrates the **energy compaction property** of the DCT:

$$\text{EnergyConcentration} = \frac{\sum_{k \in \text{kept}} |X_k|^2}{\sum_{\forall k} |X_k|^2} \approx 95\% \quad \text{with only 5\% of coefficients.}$$

Implications for JPEG and Compression Systems

1. High Compression Efficiency:

DCT concentrates most of the signal energy into a small number of low-frequency coefficients, allowing aggressive compression with minimal perceptual loss.

2. Quantization Strategy:

JPEG quantization tables allocate more bits to low-frequency coefficients (which carry most of the energy) and fewer bits (or zero) to high-frequency coefficients.

3. Perceptually Optimal:

The human visual system is less sensitive to high-frequency detail loss, making DCT-based compression visually efficient.

4. Progressive Transmission:

By transmitting low-frequency coefficients first, images can be displayed gradually from blurry to sharp (Progressive JPEG).

Q: Why use the logarithmic spectrum $\log(1+|F|)$ instead of directly plotting $|F|$?

A:

The logarithmic spectrum is necessary because:

1. Dynamic Range Issue:

The DC coefficient $(0, 0)$ is typically orders of magnitude larger than other coefficients:

$$|F_{0,0}| \approx 10^5 \quad vs. \quad |F_{k_1,k_2}| \approx 10^0 - 10^2.$$

2. Visualization Problem:

When plotting $|F|$ directly, the DC component dominates the display, making all other coefficients appear nearly black and revealing little structural information.

3. Logarithmic Compression:

The transformation

$$\log(1 + |F|)$$

compresses the dynamic range:

$$\log(1 + 10^5) \approx 11.5, \quad \log(1 + 10^2) \approx 4.6.$$

This brings coefficient magnitudes into a visually comparable range.

4. Perceptually Relevant:

The logarithmic scale mimics human visual perception, which responds approximately logarithmically to intensity (Weber–Fechner law), making subtle frequency components visible.

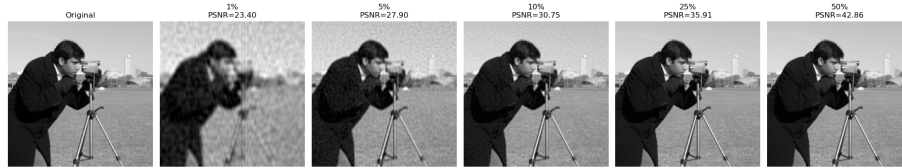


Figure 2: Image reconstruction using different percentages of DCT coefficients. The PSNR values indicate the quality of reconstruction for each percentage.

Part 4: Filtering in the Transform Domain

Q: In the output of low-pass filters, what visual changes do you observe? Why does quality change with increasing radius r ?

A:

Visual Changes for Low-Pass Filters

Radius r	Visual Effect
$r = N/8$	Severe blurring, loss of fine details, edges become diffuse, image appears smooth and soft.
$r = N/4$	Moderate blurring, some details preserved, edges slightly softened, texture reduced.

Why Quality Changes with Increasing Radius

- Smaller r preserves only very low frequencies: severe smoothing and loss of detail.
- Larger r preserves more frequency components: sharper edges and more image detail are retained.

The basis images corresponding to low frequencies have **slow spatial variations** (smooth gradients), whereas high-frequency basis images capture **rapid spatial variations** (edges and textures). Removing the latter results in blurring.

Q: What features of the image (edges, textures) does the high-pass filter output reveal?

A:

The high-pass filter output reveals:

- **Edges:** Sharp intensity transitions where objects meet.
- **Textures:** Fine patterns and surface details.
- **Noise:** High-frequency components may include image noise.
- **Contours:** Outlines of objects and structural boundaries.

The resulting image appears as a **dark background with bright outlines** at object boundaries, similar to an edge-detection result. This is because:

$$High - Pass = Original - Low - Pass.$$

Edges correspond to high-frequency content that is removed by low-pass filtering.

Q: By referring to the basis images generated in Part 1, analyze the reason for this visual behavior.

A:

Basis Image Type	Characteristics	Filtering Effect
Low Frequency (k_1, k_2 small)	Smooth gradients, gradual variations	Preserved by LP, removed by HP
High Frequency (k_1, k_2 large)	Rapid oscillations, checkerboard patterns	Removed by LP, preserved by HP

The basis images demonstrate that:

- Low-frequency basis images (top-left of the grid) represent the **gross structure** and overall intensity distribution of the image.
- High-frequency basis images (bottom-right of the grid) represent **fine details**, edges, and rapid intensity changes.

Low-Pass Filtering:

When we retain only coefficients satisfying

$$\sqrt{k_1^2 + k_2^2} \leq r,$$

we sum only the low-frequency basis images. The resulting image is dominated by smooth variations, producing the observed blurring effect.

High-Pass Filtering:

When we retain only coefficients satisfying

$$\sqrt{k_1^2 + k_2^2} > r,$$

we sum only the high-frequency basis images. This reveals edges and rapid transitions because these features are encoded in the high-frequency components.

The radial threshold r creates a circular frequency cutoff in the 2D frequency domain:

$$k_1^2 + k_2^2 \leq r^2 \quad \Rightarrow \quad \text{Low frequencies (smooth regions),}$$

$$k_1^2 + k_2^2 > r^2 \quad \Rightarrow \quad \text{High frequencies (edges and details).}$$

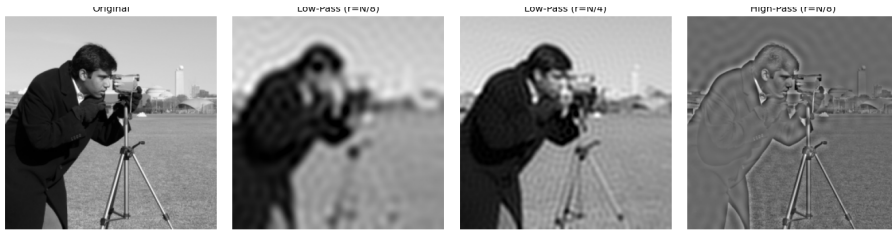


Figure 3: Filtering results in the transform domain: (a) Original image, (b) Low-pass filter with $r = N/8$, (c) Low-pass filter with $r = N/4$, (d) High-pass filter with $r = N/8$.